

HOTELPRO MANAGEMENT SUITE

WEB PROGRAMMING PROJECT

A MINI-PROJECT REPORT

Submitted by

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IN

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RAJALAKSHMI ENGINEERING COLLEGE, CHENNAI

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BONAFIDE CERTIFICATE

Certified that the project entitled "**HOTELPRO MANAGEMENT SUITE**", submitted as a CS23531 – Web Programming Project, is the bonafide work of **RAFIQ ROSHAN M - (2116241001182)**, who carried out the work under my supervision.

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INTERNAL EXAMINER **EXTERNAL EXAMINER**

ABSTRACT

Managing room reservations, guest records, billing and housekeeping is a daily requirement in every hotel, lodge and guest house. This project presents a web-based hotel management application named HotelPro Management Suite that brings these operations together in a single, lightweight and responsive interface. The system shows a live dashboard with occupancy figures, a colour-coded room availability board, and a booking form that checks dates, selects an available room and calculates the tariff and GST automatically. Guests can be checked in and checked out with one click, and itemised invoices are generated with restaurant, laundry and room-service charges.

The application is built with HTML for structure, CSS for styling and JavaScript for behaviour. Room, booking and housekeeping data are handled in the browser, and every action immediately updates the dashboard, so the front desk always sees the correct room status and double booking of a room is prevented. A light and dark theme and a mobile-friendly layout improve usability across desktops, tablets and phones. No installation, server or database is required, so the application can be hosted free of cost on platforms such as GitHub Pages.

By combining a clean user interface with reliable booking and billing logic, the system aims to be simple, fast and affordable for small hotels. The proposed system also provides a foundation for future improvements such as online payment gateways, guest login, SMS and e-mail confirmations, integration with online travel agencies, and cloud-based storage of booking history.

ACKNOWLEDGEMENT

First, we thank the almighty God for the successful completion of the project. Our sincere thanks to our chairman Mr. S. Meganathan, B.E., F.I.E., for his sincere endeavor in educating us in his premier institution. We would like to express our deep gratitude to our beloved Chairperson Dr. Thangam Meganathan, for her enthusiastic motivation which inspired us a lot in completing this project and Vice-Chairman Mr. Abhay Shankar Meganathan, B.E., M.S., for providing us with the requisite infrastructure.

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TABLE OF CONTENTS

S.No	Title	Page No
1	**INTRODUCTION**	**6**
1.1	Introduction	6
1.2	Scope of the Work	6
1.3	Problem Statement	6
1.4	Aim and Objectives of the Project	6
2	**SYSTEM SPECIFICATIONS**	**7**
2.1	Hardware Specifications	7
2.2	Software Specifications	7
3	**MODULE DESCRIPTION**	**8**
3.1	Home and Dashboard Module	8
3.2	Room Booking Module	8
3.3	Check-In, Check-Out and Billing Module	8
3.4	Housekeeping and Responsive UI Module	8
4	**SCREENSHOTS**	**9**
5	**CONCLUSION AND FUTURE ENHANCEMENT**	**15**
6	**REFERENCES**	**15**

LIST OF FIGURES

Figure No	Title of the Figure	Page No
4.1	Introduction page	9
4.2	Dashboard overview	10
4.3	Room availability board	10
4.4	Room booking form	11
4.5	Guest check-in and check-out	11
4.6	Billing and invoice	12
4.7	Housekeeping management	12
4.8	Dark theme	13
4.9	Mobile responsive view	14

CHAPTER 1
INTRODUCTION

1.1 INTRODUCTION

The HotelPro Management Suite is a web-based hotel management application that brings the daily operations of a hotel into one place. It allows the front desk to check room availability, create reservations, check guests in and out, generate bills and track housekeeping from a single dashboard. The whole application runs in the browser and requires no installation.

1.2 SCOPE OF THE WORK

The system supports a live dashboard with occupancy statistics, a colour-coded room availability board, a booking form with automatic tariff and GST calculation, guest check-in and check-out, invoice generation, housekeeping status tracking, a light and dark theme, and a responsive layout for desktops, tablets and phones. It is intended for small hotels, lodges and guest houses that need simple, affordable software.

1.3 PROBLEM STATEMENT

Many small hotels still depend on paper registers and spreadsheets to record bookings, which leads to double bookings, errors in bills and delays at the front desk. Commercial hotel software is often expensive and needs installation and training. A single, free and easy-to-use web portal that handles reservations, check-in and check-out, billing and housekeeping can reduce manual work and improve guest service.

1.4 AIM AND OBJECTIVES OF THE PROJECT

The project aims to design and develop an accurate, responsive and user-friendly web application for hotel management. The objectives are:

- to display room availability and hotel occupancy on a live dashboard;
- to implement a booking form with date validation and automatic GST billing;
- to manage guest check-in and check-out and prevent double booking of rooms;
- to generate itemised invoices with restaurant, laundry and room-service charges; and
- to provide a responsive, light and dark interface hostable on GitHub Pages.

CHAPTER 2
SYSTEM SPECIFICATIONS

2.1 HARDWARE SPECIFICATIONS

Processor	Intel i5
Memory	8 GB minimum
Storage	1 TB minimum

2.2 SOFTWARE SPECIFICATIONS

Operating system	Windows 10
Front-end	HTML5, CSS3, JavaScript (ES6)
Web browser	Google Chrome / Microsoft Edge / Firefox
Code editor	Visual Studio Code
Version control and hosting	Git and GitHub Pages
Data storage	Browser LocalStorage (no server required)

CHAPTER 3
MODULE DESCRIPTION

3.1 HOME AND DASHBOARD MODULE

The home module is the landing page of the application. It displays the welcome banner, a navigation bar and four cards that lead to Room Booking, Guest Check-In and Check-Out, Billing and Housekeeping. The dashboard shows total, occupied, available and cleaning rooms, today's arrivals, revenue and a seven-day occupancy chart. Navigation is handled in JavaScript by showing only the selected section, so switching between pages is instant and does not reload the page.

3.2 ROOM BOOKING MODULE

This module shows every room as a colour-coded tile (available, occupied, cleaning or maintenance) and allows filtering by Standard, Deluxe and Suite rooms. The booking form accepts the guest name, mobile number, e-mail, number of guests, dates and room type, and validates every field. Only available rooms of the chosen type are offered, and the number of nights, tariff, 12% GST and total amount are calculated instantly before the booking is confirmed.

3.3 CHECK-IN, CHECK-OUT AND BILLING MODULE

This module lists all bookings with a status of Reserved, Checked-In or Checked-Out. The Check-In button marks the room as occupied, and the Check-Out button releases the room and sends it for cleaning. For billing, the receptionist selects a guest, adds restaurant, laundry and room-service charges, and the system produces an itemised tax invoice with GST and the grand total, with a choice of UPI, card or cash payment.

3.4 HOUSEKEEPING AND RESPONSIVE UI MODULE

The housekeeping module lists rooms as Dirty, In Progress, Inspection, Clean or Maintenance with the assigned staff member and priority, and one button moves each room to the next stage so it becomes available again. A light and dark theme is provided using CSS variables switched by a single class on the page body, and flexbox, grid layouts and media queries adapt the interface to mobile screens.

CHAPTER 4
SCREENSHOTS

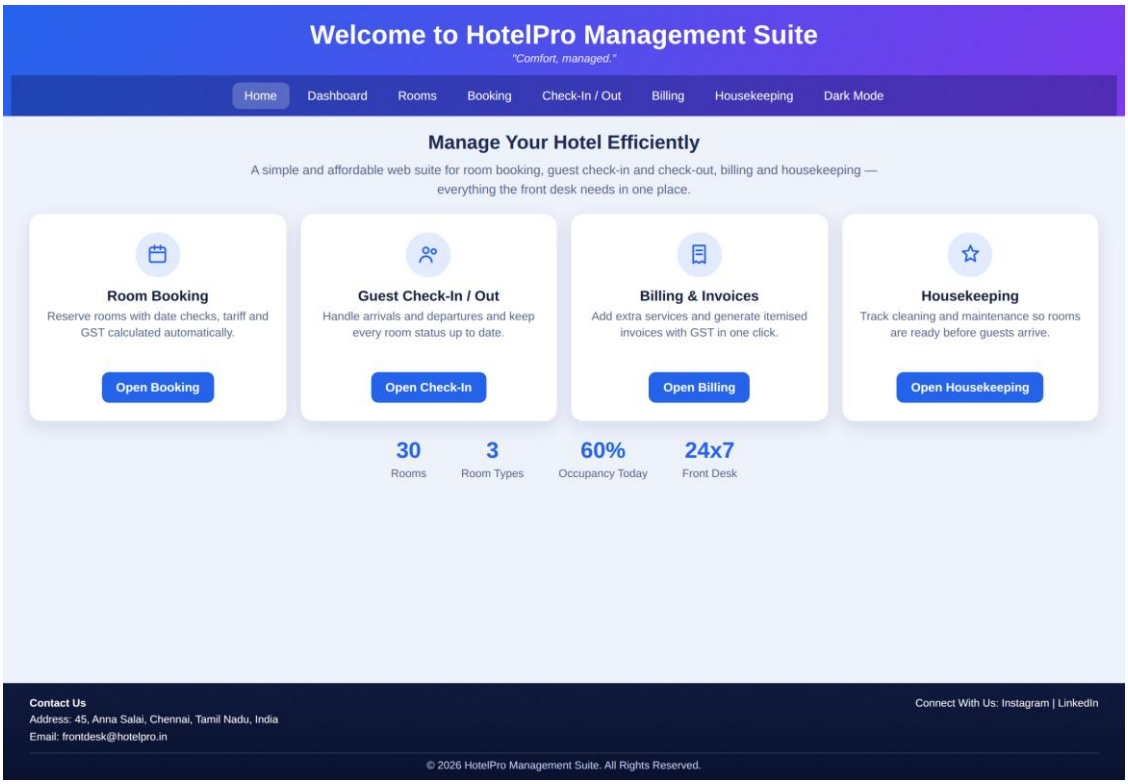


Figure 4.1: Introduction page

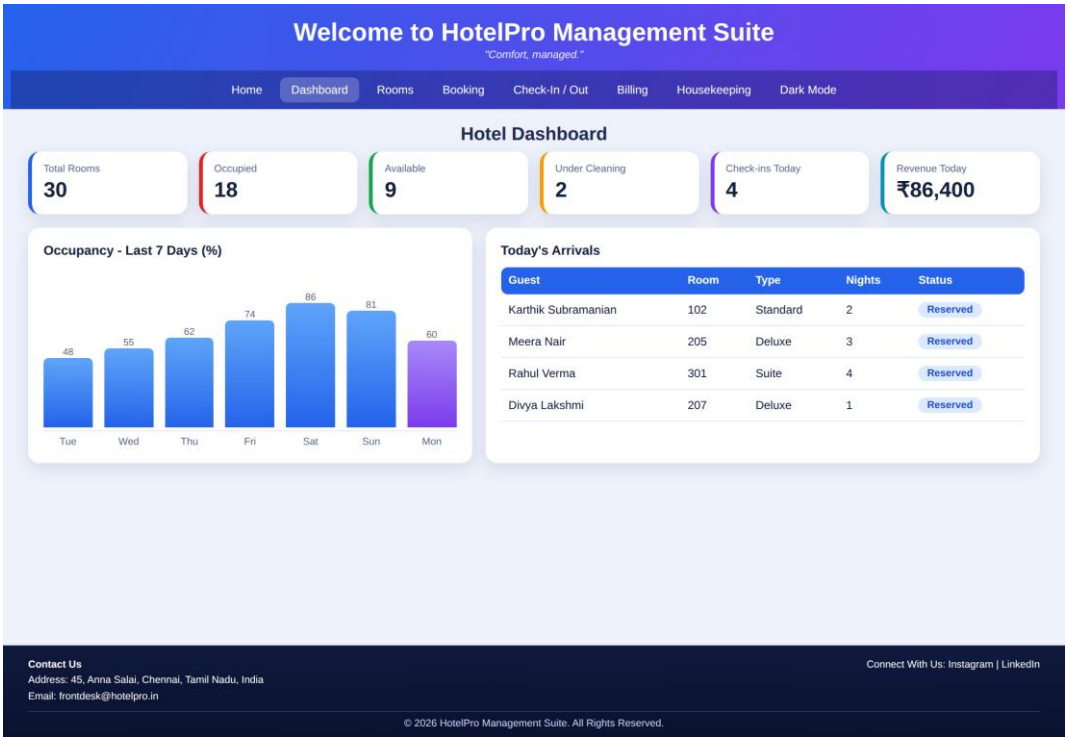


Figure 4.2: Dashboard overview

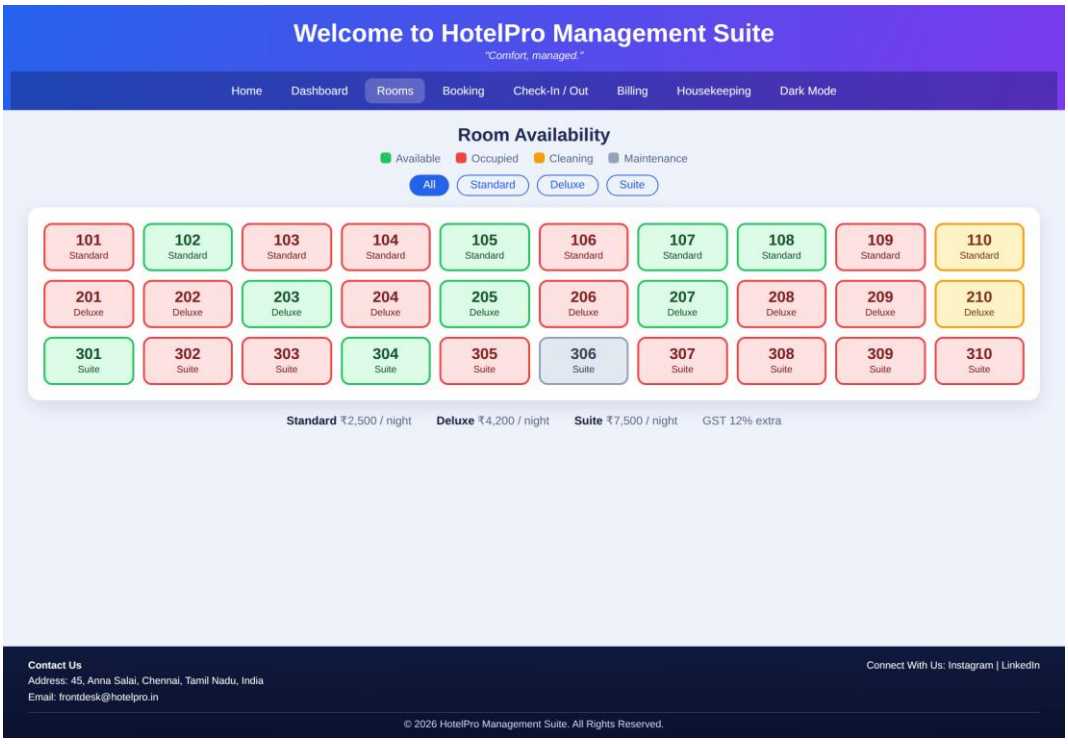


Figure 4.3: Room availability board

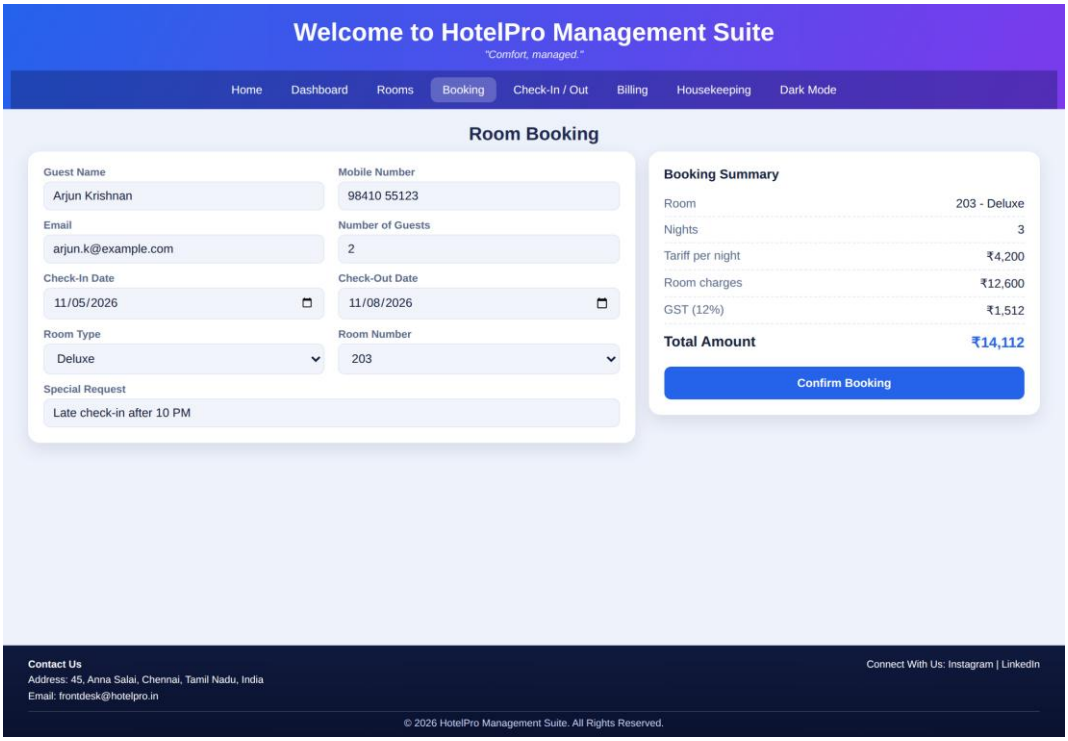


Figure 4.4: Room booking form

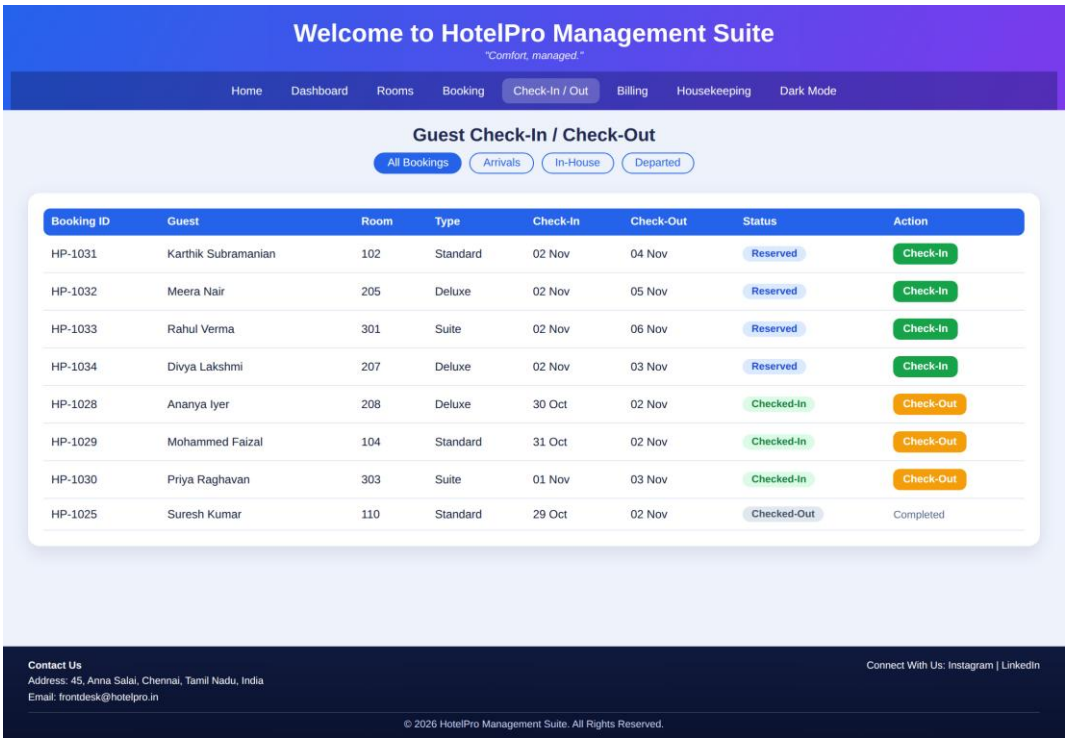


Figure 4.5: Guest check-in and check-out

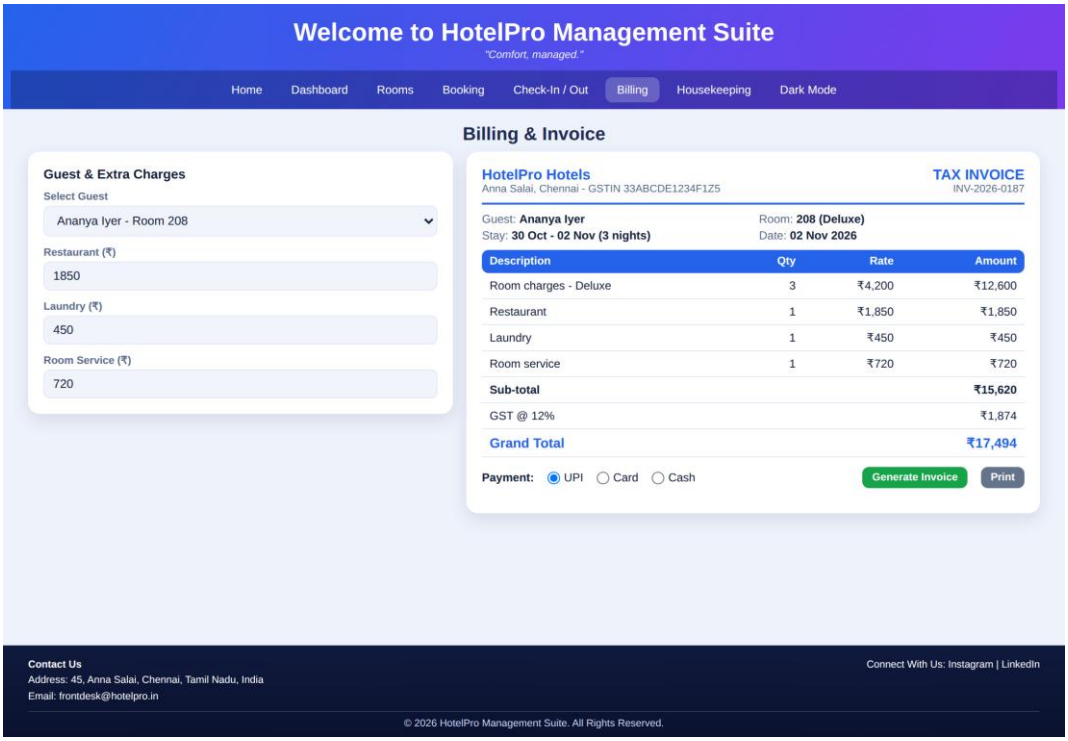


Figure 4.6: Billing and invoice

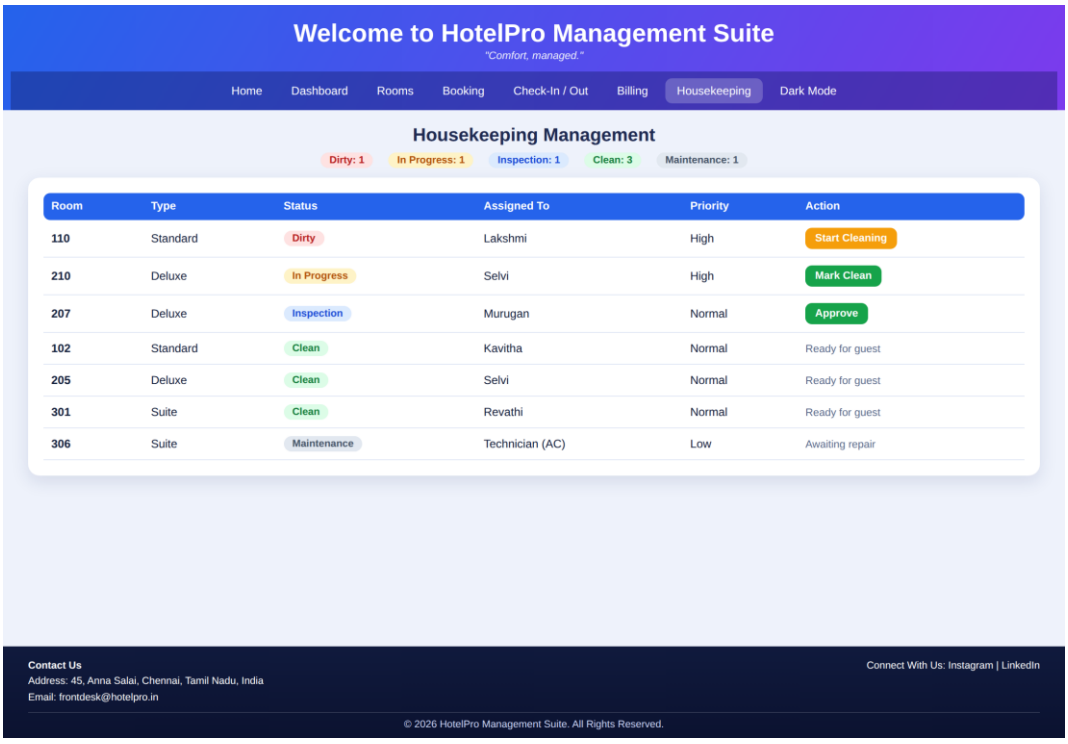


Figure 4.7: Housekeeping management

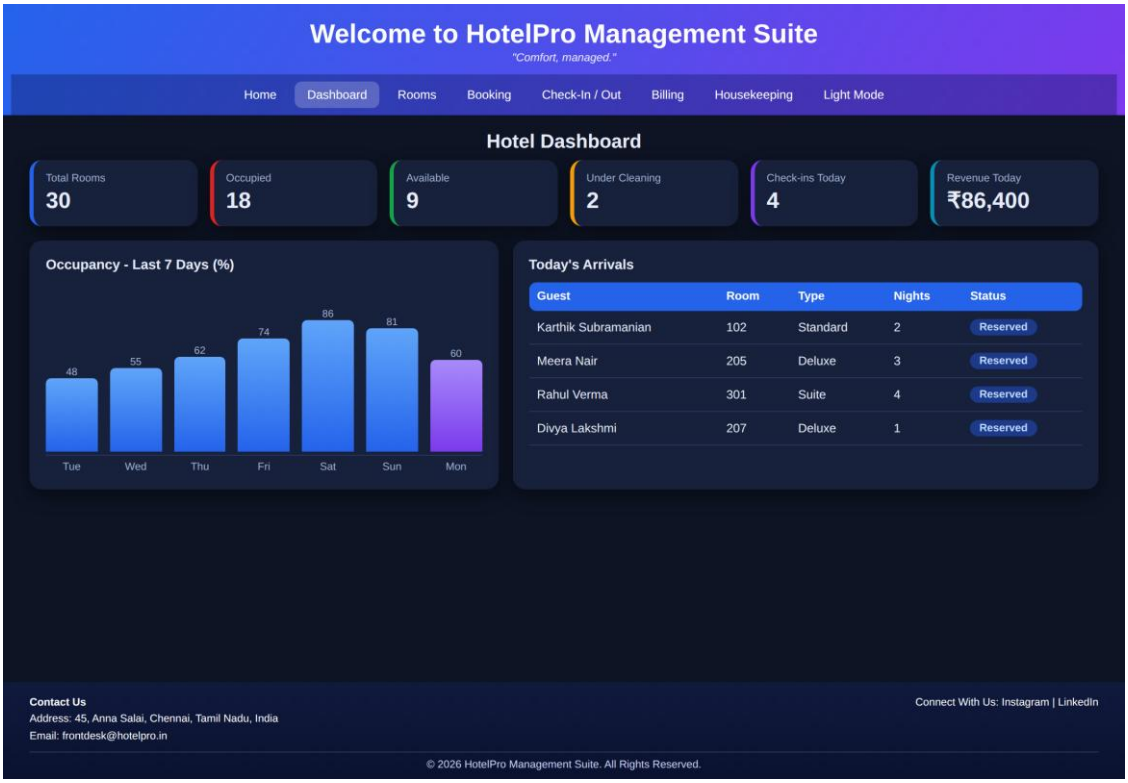


Figure 4.8: Dark theme

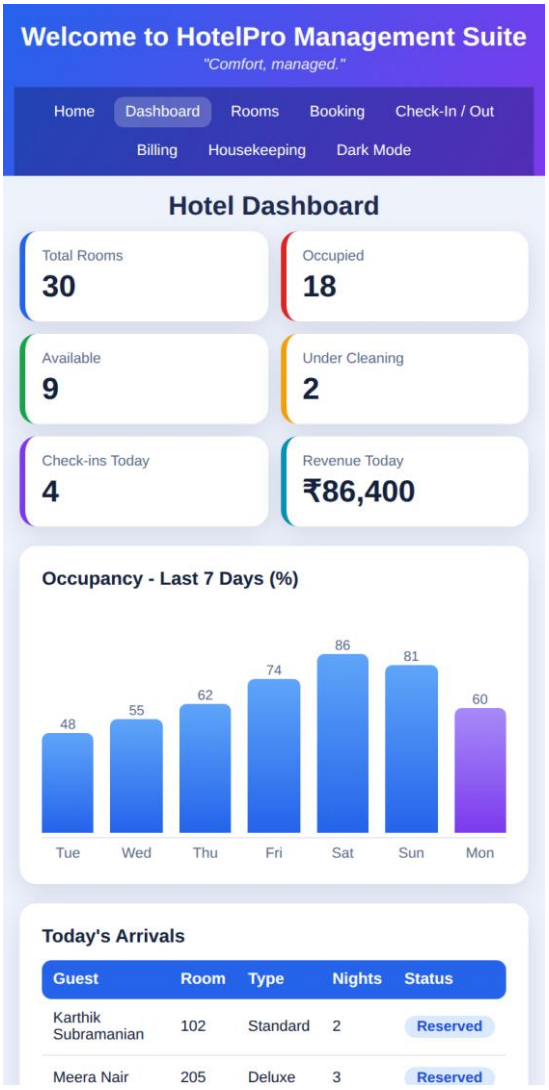


Figure 4.9: Mobile responsive view

CHAPTER 5
CONCLUSION AND FUTURE ENHANCEMENT

The HotelPro Management Suite provides a simple, accurate and accessible way to manage the daily operations of a hotel. With its dashboard, room availability board, automatic GST billing and check-in and check-out handling, it reduces paperwork and prevents double bookings. Since it runs entirely in the browser, it is fast, free to host and works on all modern devices.

Future enhancements may include secure staff login with role-based access, online payment gateway integration, SMS and e-mail booking confirmations, a guest self-booking portal, integration with travel agencies, exporting reports to PDF and Excel, offline support as a progressive web app, and cloud-based storage of booking history.

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